

ServiceNow Academic Project Report

Topic: Script-Controlled Access Control List (ACL) – Restricting Record Access Based on Field Values

Project Title	Script-Controlled ACL Implementation in ServiceNow
Platform	ServiceNow Developer Instance (PDI)
Security Module	System Security > Access Control (ACL)
Target Table	u_institution_details

1. Project Objective

The goal of this project is to create a custom table named `u_institution_details` in ServiceNow, assign user roles, and apply script-controlled Access Control Lists (ACLs) to restrict access to records based on field values and user impersonation testing.

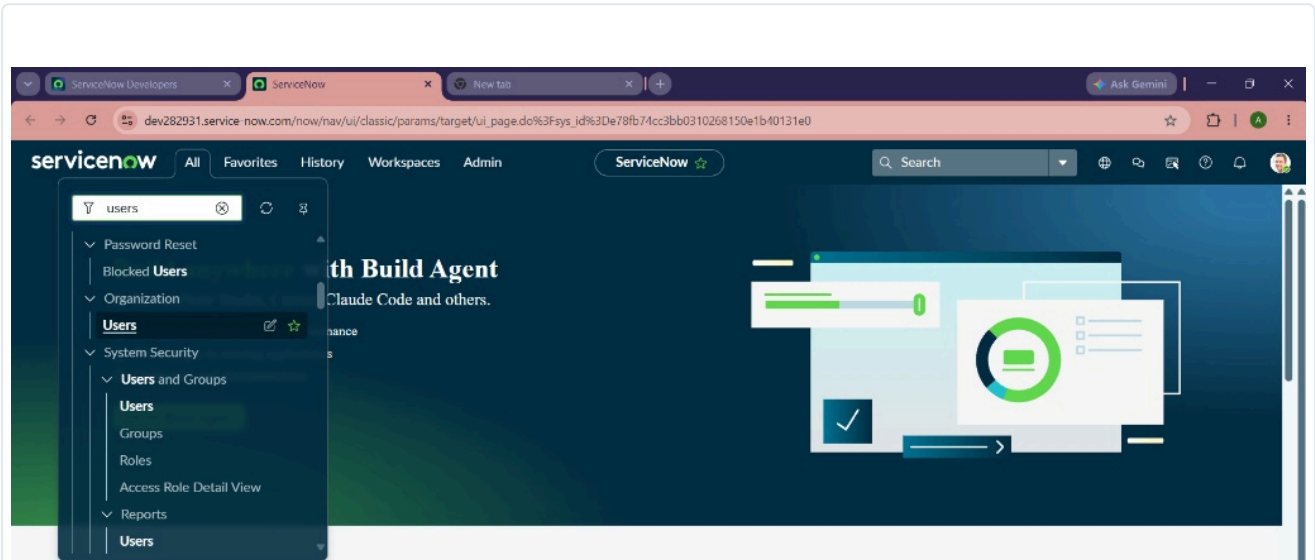
2. Workflow Steps Overview

The project follows these main steps to configure and verify security controls:

- **Step 1:** Navigate to System Security > Users.
- **Step 2:** Create a new user profile (EEE user).
- **Step 3:** Open System Security > Roles.
- **Step 4:** Create a custom security role (BB1).
- **Step 5:** Navigate to System Definition > Tables.
- **Step 6:** Create the custom table `u_institution_details`.
- **Step 7 & 8:** Elevate security privileges to `security_admin`.
- **Step 9 & 10:** Open Access Control (ACL) rules and review table permissions.
- **Step 11 & 12:** Impersonate EEE user and test record access restrictions.

Step 1: Navigating to Users Module

In this step, we opened the All Applications menu in ServiceNow and searched for "users". Under **System Security > Users and Groups**, we selected **Users** to view and create user accounts.



Step 2: Creating New User Record

We created a new user profile with the User ID **EEE user**, First name **EEE**, Last name **user**, and Email **eeeuser@gmail.com**. We set the record as Active and saved it.

The screenshot displays the ServiceNow 'User - New Record' form. The browser address bar shows the URL: `dev282931.service-now.com/now/hav/ui/classic/params/target/sys_user.do%3Fsys_id%3D-1%26sys_is_list%3Dtrue%26sys_target%3Dsys_user%26sysparm_checked_items%3D%26sysparm...`. The ServiceNow navigation bar includes the logo, 'All', 'Favorites', 'History', 'Workspaces', 'Admin', and a 'User - New Record' button. A search bar is also present. A blue notification bar at the top states: 'To set up the User's password, save the record and then click Set Password.' The form fields are organized into two columns. The left column contains: 'User ID' (filled with 'EEE user'), 'First name' (filled with 'EEE'), 'Last name' (filled with 'user'), 'Title' (empty), 'Department' (empty with a search icon), 'Password needs reset' (checkbox), 'Locked out' (checkbox), 'Active' (checkbox, checked), 'Identity type' (dropdown menu showing 'Human'), and 'Internal Integration User' (checkbox). The right column contains: 'Email' (filled with 'eeeuser@gmail.com' and an email icon), 'Language' (dropdown menu showing '-- None --'), 'Calendar integration' (dropdown menu showing 'Outlook'), 'Time zone' (dropdown menu showing 'System (America/Los_Angeles)'), 'Date format' (dropdown menu showing 'System (2026-08-19)'), 'Business phone' (empty), 'Mobile phone' (empty), and 'Photo' (text 'Click to add...'). At the bottom left, there is a 'Submit' button and a 'Related Links' section with two links: 'View linked accounts' and 'View Subscriptions'. A small information icon is located at the bottom right of the form area.

User ID: EEE user

First name: EEE

Last name: user

Title:

Department:

Password needs reset: ☐

Locked out: ☐

Active: ☒

Identity type: Human

Internal Integration User: ☐

Email: eeeuser@gmail.com

Language: -- None --

Calendar integration: Outlook

Time zone: System (America/Los_Angeles)

Date format: System (2026-08-19)

Business phone:

Mobile phone:

Photo: Click to add...

Submit

Related Links

- [View linked accounts](#)
- [View Subscriptions](#)

Figure 2: User record creation for EEE user.

Step 3: Navigating to Roles Module

To create a custom role for access testing, we used the All Applications menu to search for "roles" and clicked on **System Security > Users and Groups > Roles**.

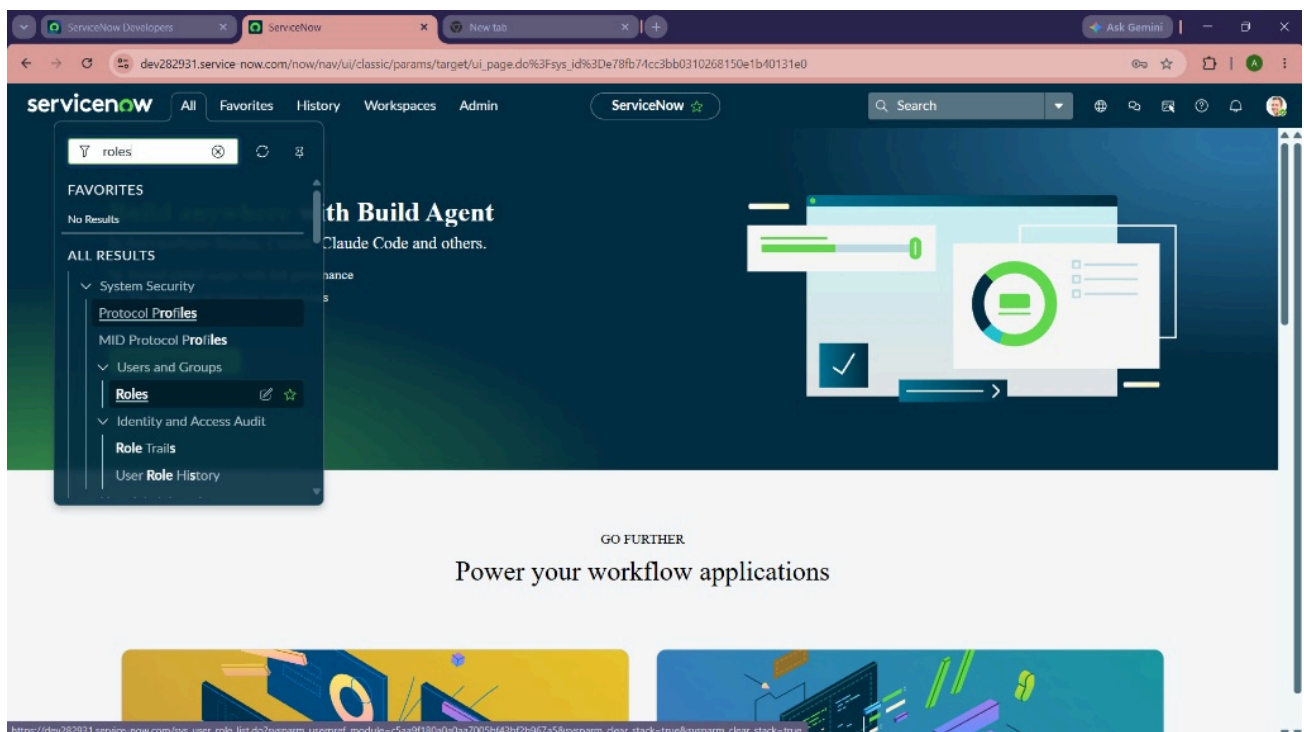


Figure 3: Searching for System Security > Roles.

Step 4: Creating Custom Role "BB1"

We clicked **New** to create a custom security role named **BB1** under the Global application scope. This role will be assigned to users who require access to our custom institution records.

The screenshot displays the ServiceNow 'Role - New Record' form. The browser address bar shows the URL: dev282931.service-now.com/now/nav/ui/classic/params/target/sys_user_role.do%3Fsys_id%3D-1%26sys_is_list%3Dtrue%26sys_target%3Dsys_user_role%26sysparm_checked_items%3D%2... The ServiceNow header includes the logo, navigation tabs (All, Favorites, History, Workspaces, Admin), and a search bar. The form title is 'Role - New Record'. The 'Name' field is labeled with a red asterisk and contains the text 'BB1'. The 'Application' dropdown menu is set to 'Global'. The 'Elevated privilege' checkbox is unchecked. The 'Description' field is empty. A 'Submit' button is located at the bottom left of the form area.

Figure 4: Creating the BB1 custom security role.

Step 5: Navigating to System Definition > Tables

Next, we prepared to create our custom database table. We searched for "tables" in the navigation filter and selected **System Definition > Tables**.

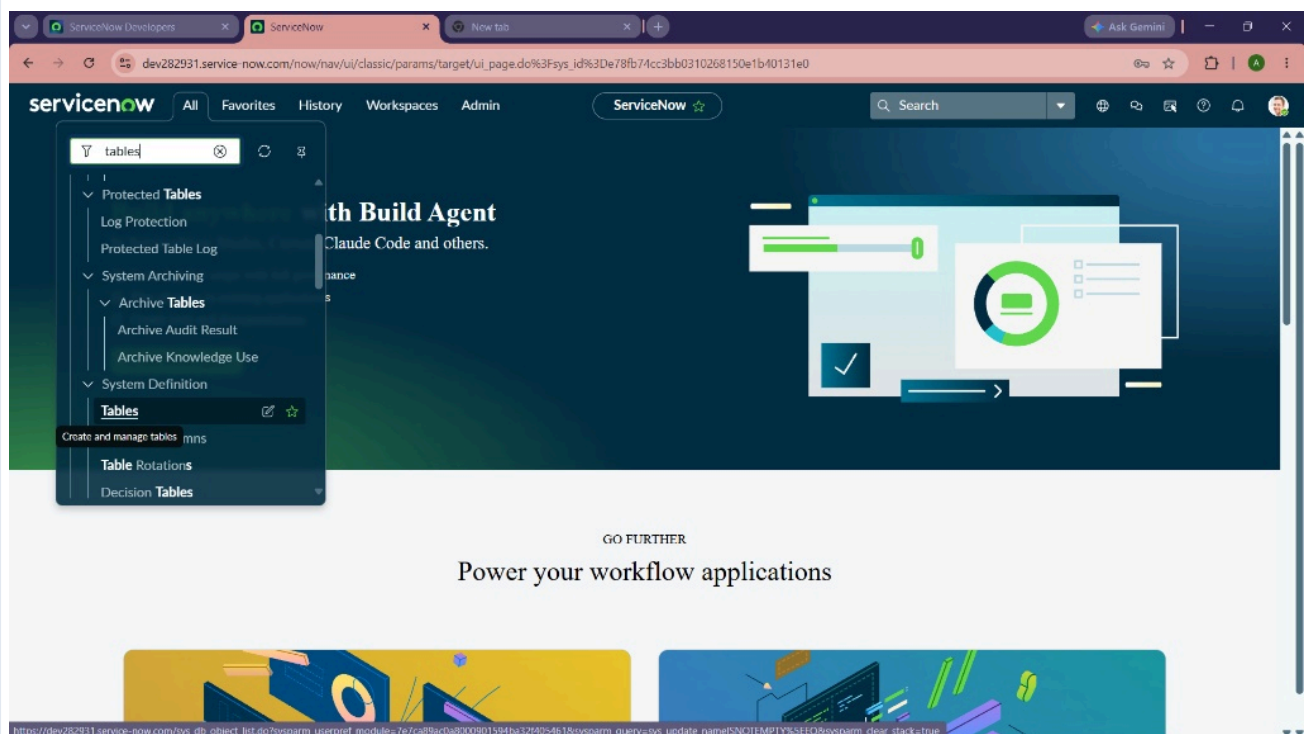


Figure 5: Accessing System Definition > Tables.

Step 6: Creating Table "u_institution_details"

We created a new table with the Label **Institution Details**, which automatically set the Name to **u_institution_details**. We enabled *Create module* so it appears in the navigation menu.

The screenshot shows the ServiceNow 'Table - New Record' form. The 'Label' field is set to 'Institution Details' and the 'Name' field is set to 'u_institution_details'. The 'Application' is set to 'Global'. The 'Create module' checkbox is checked, and the 'Create mobile module' checkbox is also checked. The 'Add module to menu' dropdown is set to '-- Create new --', and the 'New menu name' field is set to 'Institution Details'. The 'Remote Table' checkbox is unchecked. Below the form, there is a 'Columns' tab with a 'Table Columns' section and a 'Dictionary Entries' table.

Column label	Type	Reference	Max length	Default value	Display
Insert a new row...					

Figure 6: Defining the u_institution_details table fields and navigation module.

Step 7: Selecting "Elevate role" Option

* Note: The screenshot below shows the elevated menu panel highlighted during selection.

To modify or create security rules (ACLs), standard administrator rights are not sufficient. We clicked on the System Administrator profile icon in the top right corner and selected **Elevate role**.

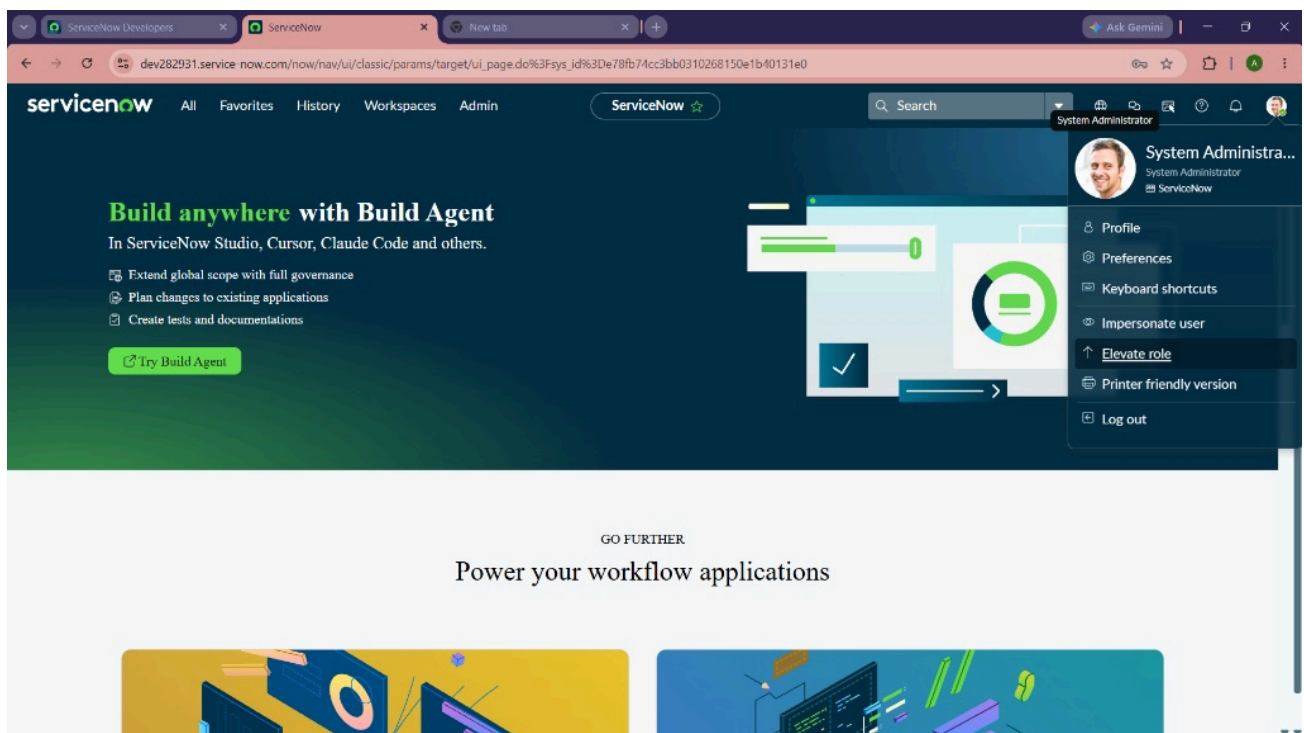


Figure 7: Opening the profile menu to elevate system privileges.

Step 8: Elevating Privileges to security_admin

In the popup modal, we checked the **security_admin** checkbox and clicked **Update**. This grants temporary elevated access required to modify Access Control Lists.

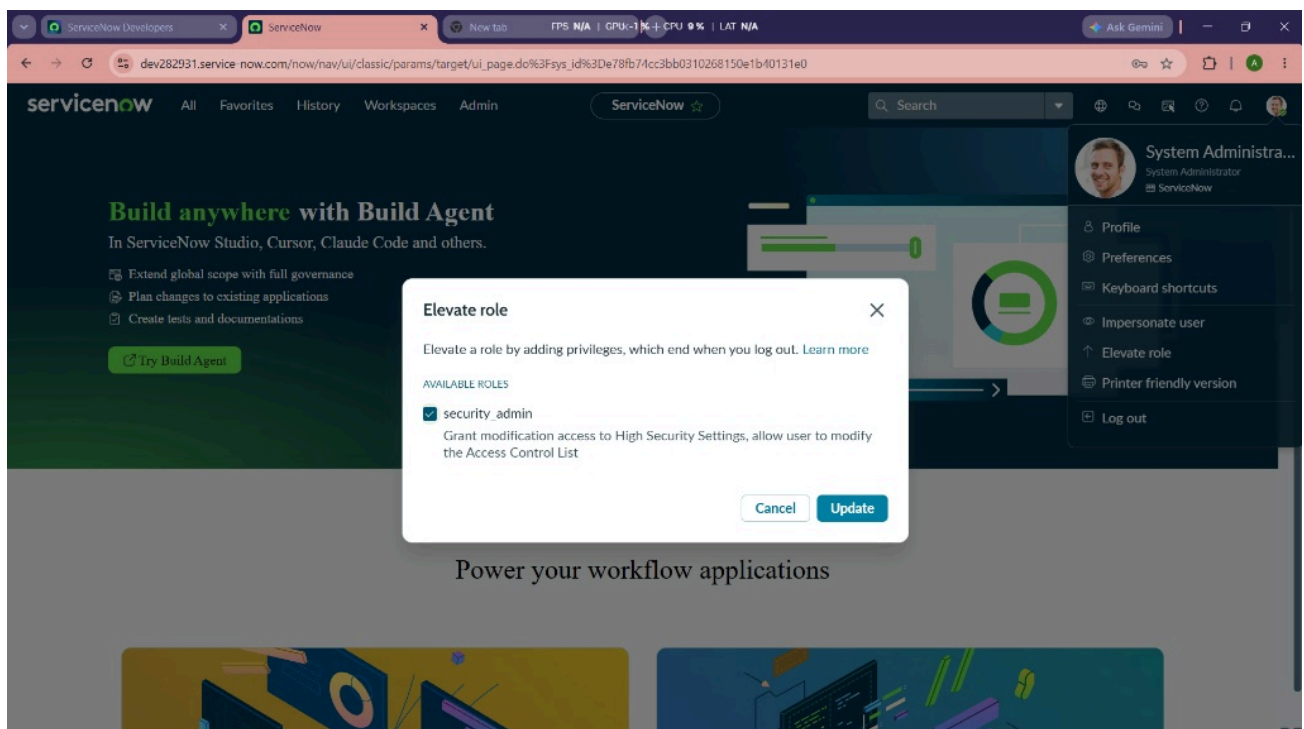


Figure 8: Checking security_admin role and clicking Update.

Step 9: Opening Access Control (ACL) Module

With elevated security_admin privileges active, we searched for "acl" in the All menu and clicked on **System Security > Access Control (ACL)**.

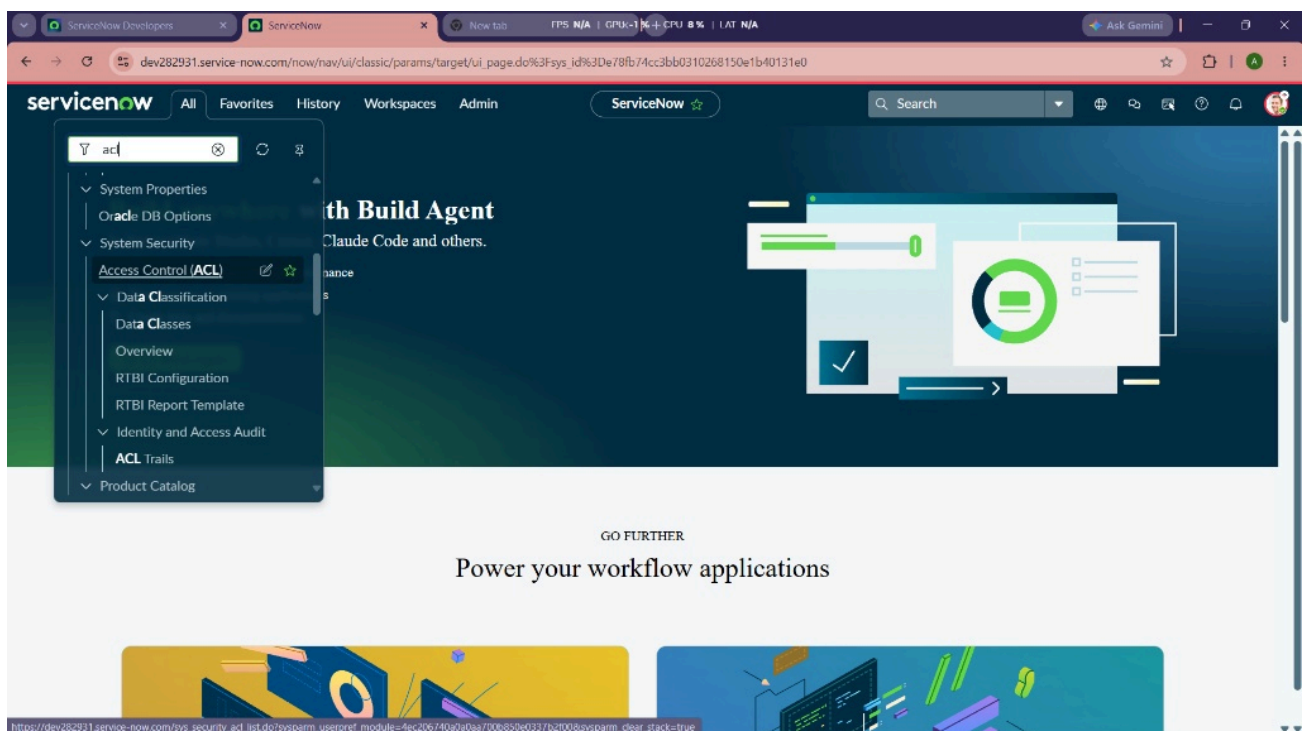
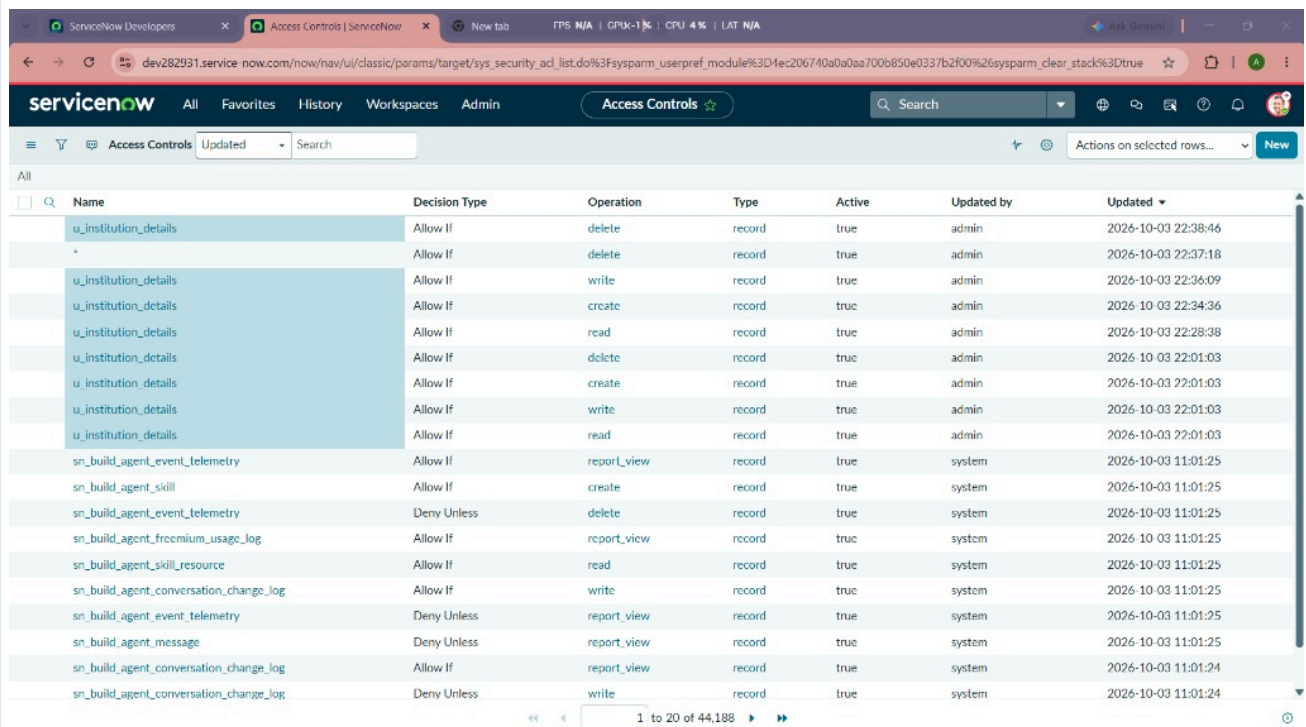


Figure 9: Accessing System Security > Access Control (ACL).

Step 10: Viewing Access Control List Rules

We filtered the ACL list to view all default access rules for `u_institution_details`. The list shows auto-generated rules for **create**, **read**, **write**, and **delete** operations.



The screenshot shows the ServiceNow Access Controls interface. The table displays ACL rules for the `u_institution_details` entity. The rules are filtered to show only those for the `u_institution_details` entity. The table columns are: Name, Decision Type, Operation, Type, Active, Updated by, and Updated. The rules are sorted by Updated date, showing the most recent rules at the top.

Name	Decision Type	Operation	Type	Active	Updated by	Updated
u_institution_details	Allow If	delete	record	true	admin	2026-10-03 22:38:46
*	Allow If	delete	record	true	admin	2026-10-03 22:37:18
u_institution_details	Allow If	write	record	true	admin	2026-10-03 22:36:09
u_institution_details	Allow If	create	record	true	admin	2026-10-03 22:34:36
u_institution_details	Allow If	read	record	true	admin	2026-10-03 22:28:38
u_institution_details	Allow If	delete	record	true	admin	2026-10-03 22:01:03
u_institution_details	Allow If	create	record	true	admin	2026-10-03 22:01:03
u_institution_details	Allow If	write	record	true	admin	2026-10-03 22:01:03
u_institution_details	Allow If	read	record	true	admin	2026-10-03 22:01:03
sn_build_agent_event_telemetry	Allow If	report_view	record	true	system	2026-10-03 11:01:25
sn_build_agent_skill	Allow If	create	record	true	system	2026-10-03 11:01:25
sn_build_agent_event_telemetry	Deny Unless	delete	record	true	system	2026-10-03 11:01:25
sn_build_agent_freemium_usage_log	Allow If	report_view	record	true	system	2026-10-03 11:01:25
sn_build_agent_skill_resource	Allow If	read	record	true	system	2026-10-03 11:01:25
sn_build_agent_conversation_change_log	Allow If	write	record	true	system	2026-10-03 11:01:25
sn_build_agent_event_telemetry	Deny Unless	report_view	record	true	system	2026-10-03 11:01:25
sn_build_agent_message	Deny Unless	report_view	record	true	system	2026-10-03 11:01:25
sn_build_agent_conversation_change_log	Allow If	report_view	record	true	system	2026-10-03 11:01:24
sn_build_agent_conversation_change_log	Deny Unless	write	record	true	system	2026-10-03 11:01:24

Figure 10: Reviewing existing ACL rules for `u_institution_details`.

Step 11: Impersonating "EEE User" for Access Testing

To test our script-controlled access rule, we impersonated our test user. We clicked the profile menu, selected **Impersonate user**, searched for **EEE User**, and clicked **Impersonate user**.



Step 11 Screenshot

Figure 11: Impersonating EEE User to test security rules.

Step 12: Testing Access via Navigation Search

While impersonating EEE User, we searched for `tution_details.list` in the navigator. The system returned **No Results** because EEE User lacks the required permissions/roles according to the ACL rules.



Step 12 Screenshot

Figure 12: Navigator showing "No Results" due to restricted access.

Step 13: Verifying Record Access in Institution Details List

Finally, when navigating directly to the **Institution Details** list, the user can view authorized records matching field values (such as Branch = EEE, Faculty Name = Alejandra Prenatt, Student Name = Allan Schwantd, Roll Number = STU200).



Step 13 Screenshot

Figure 13: Authorized record displayed in Institution Details list view.

Project Conclusion

In this project, we successfully created a custom table (u_institution_details), configured custom roles (BB1), created test users (EEE user), elevated system privileges to security_admin, and validated script-controlled ACL behavior through user impersonation testing.
